

RETAILPULSE 360

Sales, Profit & Customer Intelligence System

MainCrafts SkillSprint - Data Science & Data Analytics Using Python

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Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Google Colab

1. Problem Statement

A growing retail company holds thousands of transactions but does not clearly understand what drives its sales and profit. Management wants to know which products, regions, customer segments and discount levels create value, and where money is being lost.

2. Objective

Build RETAILPULSE 360, a Python analytics workflow that follows Understand, Clean, Analyze, Visualize, Discover, Explain, Recommend, and turns raw transactions into business recommendations that a non-technical manager can use.

3. Dataset

Global Superstore retail transaction data: 51,290 rows and 27 columns covering 2011-2014, with 25,035 orders, 4,873 customers, 147 countries, 13 regions, 3 categories and 17 sub-categories. Key fields: Order Date, Ship Date, Customer, Segment, Region, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Ship Mode, Order Priority.

4. Methodology

- Loaded the CSV in Google Colab with Pandas and inspected structure, data types and summary statistics.
- Ran data-quality checks (missing values, duplicates, data types, unique counts).
- Cleaned the data: converted dates to datetime and rounded Discount to 2 decimals to remove floating-point duplicates.
- Engineered features: Year, Month, Month Name, Order Month, Shipping Days, Profit Margin (zero-sales safe).
- Analysed KPIs, sales, profitability, products, customers, segments, regions, time series, discount effect and shipping using groupby aggregations, then visualised results with Matplotlib and Seaborn.

5. Data Cleaning Summary

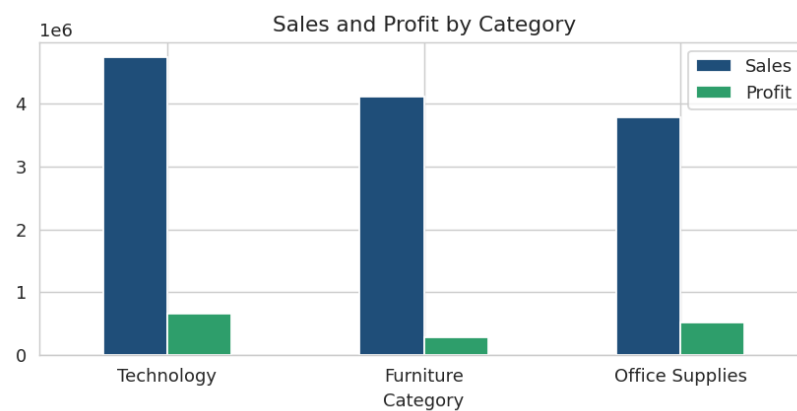
Check	Result
Total records / columns	51,290 / 27
Missing values	0
Duplicate rows	0 (none removed)
Date columns	Order Date, Ship Date converted from text to datetime (0 invalid dates)
Other fix	Discount rounded to 2 decimals so the same discount level is not split into two groups
Note	The Chinese-named record-count column is constant (value 1) and was left unused

6. Key Performance Indicators

KPI	Value
Total Sales	12,642,905
Total Profit	1,467,457
Profit Margin	11.61%
Total Orders	25,035
Total Customers	4,873
Units Sold	178,312
Average Order Value	505.01

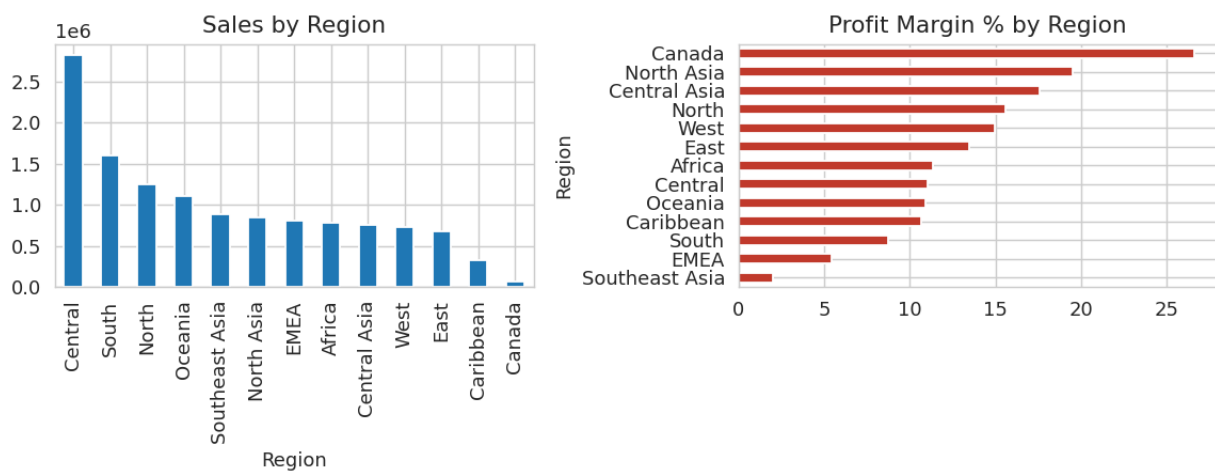
7. Analysis and Visualisations

7.1 Category



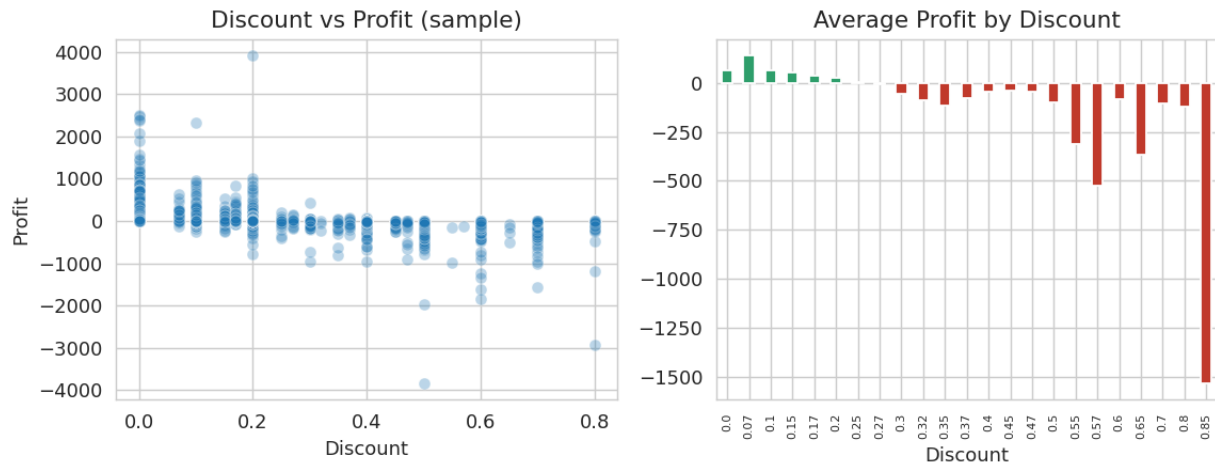
Technology leads in both sales (4.74M) and profit (663.8K, 14.0% margin). Furniture has 4.11M in sales but only a 6.9% margin, the weakest of the three. Office Supplies earns a 13.7% margin.

7.2 Regions



Central is the largest region (2.82M, 22.3% of sales). High sales do not guarantee high profit: South has the second-highest sales but only an 8.8% margin, Southeast Asia (2.0%) and EMEA (5.5%) are weak, while Canada has the best margin (26.6%) on very small sales.

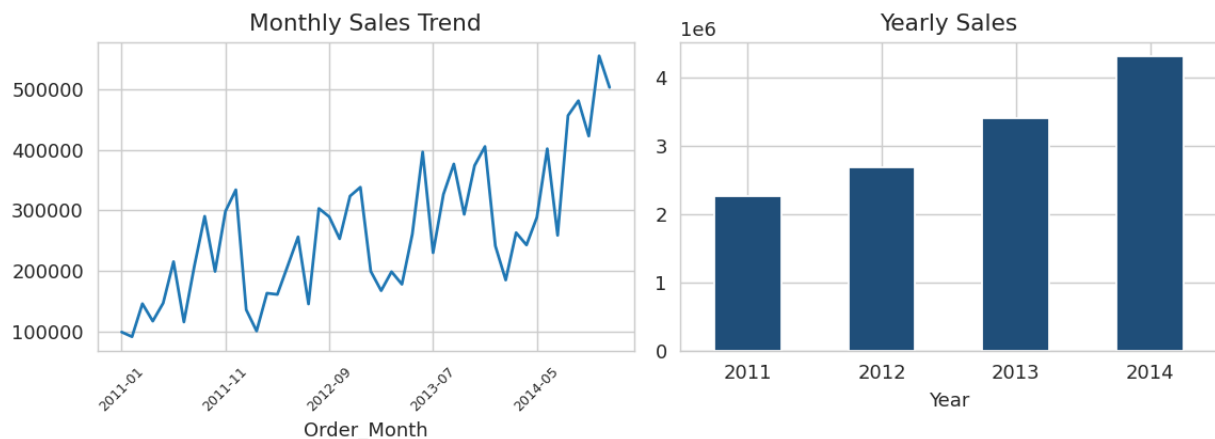
7.3 Discount and Profit



Discount band	Avg profit per line	Lines	Total profit
0%	62.05	29,470	1,828,672
0-20%	43.00	10,533	452,872
20-30%	-22.20	926	-20,560
30-50%	-61.55	6,189	-380,945
50%+	-98.89	4,172	-412,582

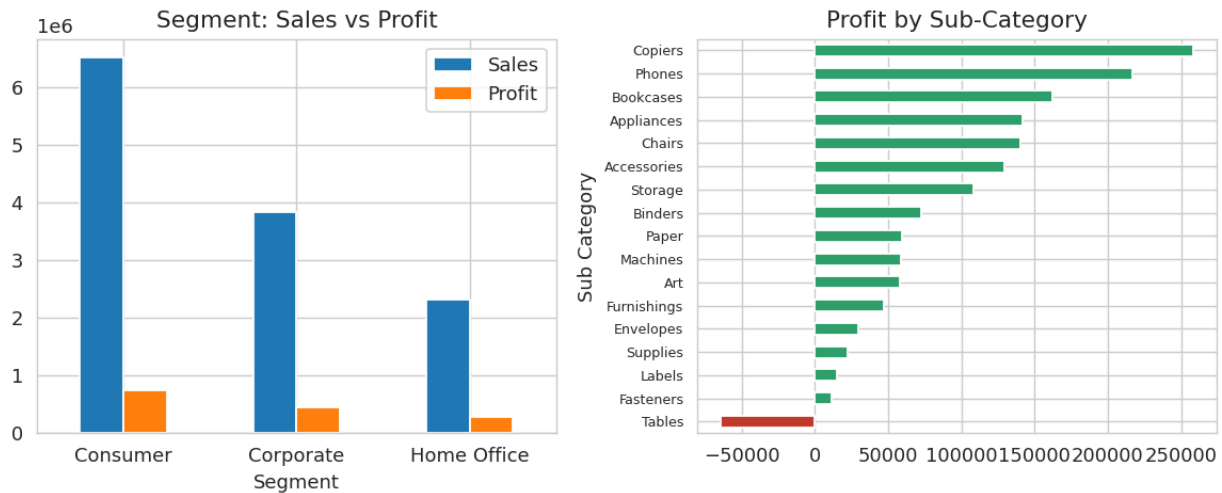
Correlation between discount and profit is -0.32. Average profit turns negative above roughly 27% discount. This is an association, not proof of causation: product mix, category and region also affect profit, and the reason a discount was given is not in the data.

7.4 Time Trend



Sales grew every year: 2.26M (2011), 2.68M (2012), 3.41M (2013), 4.30M (2014), i.e. +18.5%, +27.2% and +26.3%. The highest month was November 2014 (555K); September, November and December are the strongest calendar months.

7.5 Segments and Sub-Categories



Segment	Sales	Profit	Margin	Customers
Consumer	6,508,141	749,240	11.51%	2,509.0
Corporate	3,824,808	441,208	11.54%	1,457.0
Home Office	2,309,956	277,009	11.99%	907.0

Consumer is the largest segment by sales, profit and customers, while Home Office has the highest margin (11.99%). Tables is the only loss-making sub-category (-64,083).

7.6 Products, Customers and Shipping

- 677 of 3,788 products are loss-making. The top-10 loss makers (Cubify 3D printers, Lexmark laser printer, Bevis tables and others) lost about 40,050 on 143,675 of sales, with an average discount of 24.2% versus 14.3% overall.
- Top customer by sales: Sean Miller (25,042). Top by profit: Tamara Chand (8,981). Only 5 of the top-10 sales customers are also in the top-10 profit list. The top 10% of customers contribute 31.3% of sales, and 1,058 customers are loss-making.
- Shipping margin is almost identical across modes (11.4% to 11.8%). Standard Class carries most orders (15,154) and the most profit. Average delivery time is 3.97 days, so faster shipping is not linked to higher profitability.

8. Key Findings

#	Finding	Evidence
1	Sales grew every year	2.26M to 4.30M (2011-2014)
2	Technology is the largest and most profitable category	4.74M sales, 663.8K profit
3	Furniture margin is weak; Tables lose money	6.9% margin; Tables -64,083
4	Discounts above about 30% cause losses	Corr -0.32; 30%+ lines lose 793K in total
5	Central is the core market	2.82M, 22.3% of sales
6	High sales do not equal high profit	Southeast Asia 2.0%, EMEA 5.5%, South 8.8% margin
7	677 products lose money	Top-10 loss about 40K, avg discount 24.2%
8	Many customers are unprofitable	1,058 of 4,873 negative
9	Q4 is peak season	Nov-2014 peak 555K
10	Shipping mode does not change margin	11.4%-11.8% across modes

9. Recommendations

Recommendation 1: Cap discounts

Observation and evidence: Lines with 30%+ discount lose money (avg -62 to -99 per line).

Action: Cap discounts near 20-30%; require manager approval above that.

Expected purpose: Reduce avoidable losses and lift the 11.6% margin.

Recommendation 2: Fix Furniture and Tables

Observation and evidence: Furniture margin is 6.9%; Tables lose 64,083.

Action: Raise Tables prices or renegotiate supplier cost and discounts.

Expected purpose: Make Furniture profitable.

Recommendation 3: Review loss-making products

Observation and evidence: 677 products lose money; top-10 discount averages 24.2%.

Action: Stop discounts, reprice or drop the worst products.

Expected purpose: Remove avoidable losses.

Recommendation 4: Fix weak regions

Observation and evidence: Southeast Asia 2.0%, EMEA 5.5%, South 8.8% margin; Turkey and Nigeria have the largest country losses.

Action: Review discounting and cost structure in these markets.

Expected purpose: Turn sales into profit.

Recommendation 5: Plan for Q4 and retain top customers

Observation and evidence: Nov-Dec peak; top 10% of customers give 31.3% of sales.

Action: Stock early for Q4 and offer loyalty benefits to high-profit customers.

Expected purpose: Capture peak revenue and improve retention.

10. Limitations

- The data is historical (2011-2014); future market conditions may differ.
- The analysis shows association, not causation, especially for discount versus profit.
- External factors are missing: marketing spend, competitors, returns, customer satisfaction and the reason behind each discount.

- Region and country losses may reflect product mix or shipping cost, which were not fully separated.
- Currency is not stated in the dataset, so values are shown without a currency symbol.

11. Conclusion

RetailPulse 360 shows a business that is growing quickly (sales about 1.9x between 2011 and 2014) but whose profit quality does not grow at the same pace. The main issues are heavy discounting, loss-making products and Furniture/Tables, and low-margin regions. Acting on the five recommendations can improve the 11.6% margin without reducing sales.